

A Runtime Readout Monitor for Halt-Failure in OCR Vision-Language Models: Corpus, Early-Commitment, and the Honest Precision-Recall Trade-off of a Missing Halt Signal

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Abstract

Fine-tuned OCR vision-language models (OCR-VLMs) sometimes fail to stop: instead of emitting end-of-sequence (EOS), decoding runs to the generation cap and surfaces as phantom table rows, runaway loops, and hallucinated structural markers. We take the systems view of this *halt-failure* in a 3B-parameter OCR fine-tune and argue that the deficit is best read as a *missing halt signal*: not a corrupted residual state, not mere repetition, and not a single steerable layer. Our contribution is a deployable artifact and an honest negative. We first characterize the phenomenon and corpus (56 confirmed cap-hit positives spanning 12 populated surface classes; a device-induced non-determinism that *shrinks* rather than enlarges the stable positive set; and a domain-conditional trigger gradient). We then localize *where to monitor*: a pre-registered early-detection probe shows the cap-hit outcome is linearly decodable in the first tens of generated tokens (best leave-one-out AUC 0.857 at layer 24, generation position ~ 50 , $N = 16$), so a monitor should read early at layer 24 rather than wait for the visible transition. Three converging diagnostics support the missing-halt reading: at loop onset EOS is genuinely buried (per-document median rank $\sim 12,234$, only 1 of 22 documents ever in the top 5), EOS suppression is relative not absolute (the chosen token out-scores EOS by roughly 20 logits while the EOS logit stays positive; an $N = 1$ illustration), and the loop runs on a coherent, not corrupted residual (in-loop/pre-loop residual norm 0.90 to $0.97\times$, ruling out the $\geq 3\times$ inflation that an attention-sink / extreme-token corruption account requires; $N = 3$). The systems-relevant causal conclusion is that you cannot patch one site to make it stop: six pre-registered perturbation experiments (single-direction project-out, component-resolved patching, induction-head zeroing, prompt counterfactuals, SAE-feature ablation, and a multi-layer coordinated patch) all return the FAIL/null verdict, with a norm-scaled positive control that bounds, without eliminating, the under-powered alternative. We then ship the deliverable: a runtime monitor that reads the layer-24 halt direction and boosts EOS, sitting on a genuine precision-recall trade-off with no free sweet spot. At a +6.91 boost it cuts mean runaway length 37.68% with 0 of 10 control false-positives but still leaves 5 of 13 positives at the cap; at a +30 boost it reaches 99.58% reduction but at 10 of 10 control false-positives. Grey-noise image inpainting collapses all 5 of 5 named positives to short EOS halts, motivating a vision-aware guardrail. The representational geometry that explains *why* these per-class halt signals are organized the way they are (per-class halt directions, a two-family bifurcation, count-manifold

structure, and a fine-tune signature) is treated in a companion representational-geometry paper; here we keep only the minimal mechanistic facts a practitioner needs to justify the monitor.

AI assistance disclosure. This preprint was produced with substantial AI assistance: an AI research agent drafted text, ran and logged experiments, and assembled the manuscript under the author’s direction. The author specified the research questions, designed and pre-registered the falsification criteria, audited every numeric claim against on-disk results files, and made all interpretive and editorial decisions. Numeric claims that could not be reproduced from a results file were marked unverified and excluded from the load-bearing argument rather than retained. The author takes full responsibility for the content, including any remaining errors.

1 Introduction

Autoregressive vision-language models for document understanding occasionally enter a state from which they never emit EOS. On a structured document the decoder will, for example, keep emitting `<tr><td></td></tr>` phantom rows, repeat a single filled cell value hundreds of times, or cycle a LaTeX command indefinitely, until the generation harness force-stops it at the token budget. This is a production reliability bug: it inflates latency and cost, and the runaway output surfaces to users as phantom structure and hallucinated markup. This paper takes the systems view of that bug and asks the practitioner’s questions directly: what triggers it, where and when can we detect it, can we install halting cheaply, and what does a deployed runtime fix actually buy.

Our organizing claim is that the deficit is a *missing halt signal* rather than a single localizable fault. The data point away from three tempting single-cause stories. It is not a *corrupted* residual state: the loop runs on a normal-norm, coherent residual, not the inflated attention-sink regime a corruption account predicts (Section 5). It is not *mere repetition*: the failure spans several surface classes beyond token loops, including phantom rows and structural-marker hallucination (Section 2). And it is not a *single steerable layer*: a converging series of single-direction, single-layer, and multi-layer perturbations all fail to move the token count, so you cannot patch one site to make it stop (Section 6). We are explicit about what those nulls do and do not show: a converging set of nulls at a fixed protocol power is consistent both with a genuinely distributed deficit and with an under-powered protocol, and the present data do not distinguish these. That mechanistic adjudication is out of scope for this systems paper; our claim is the deployment-relevant one, that single-site fixes do not install halting.

We make four contributions. (1) A corpus and taxonomy of halt-failure on real documents, including a device-induced non-determinism disclosure and a domain-conditional trigger gradient (Section 2). (2) An early-commitment result that says *where and when* to monitor: the cap-hit decision is linearly detectable in the first tens of generated tokens, with a layer-24 readout as the natural monitor site (Section 4). (3) A consolidated missing-halt diagnosis (EOS genuinely buried, suppression relative not absolute, residual coherent not corrupted) plus the converging perturbation nulls as the systems conclusion that single-site fixes fail (Sections 5 and 6). (4) A deployable runtime readout monitor with an honest precision-recall trade-off and a vision-sufficiency result that motivates a vision-aware guardrail (Section 8). The representational geometry that explains *why* the per-class halt signals are structured as they are is deferred to a companion representational-geometry paper.

2 The phenomenon and corpus

2.1 Failure taxonomy

Halt-failure is not monolithic. On real documents it surfaces as several distinct *surface classes*: filled-cell repetition (a single cell value repeated), bare-word repetition, LaTeX math or command loops, LaTeX structure loops, HTML phantom rows, tag spam, punctuation runs, and constant-value repetition, among others. These are surface symptoms of a shared underlying failure to terminate. One of our central method choices (Section 3) is to probe for halt directions *per class* rather than assume one shared representation.

2.2 Corpus construction and a non-determinism disclosure

Real public documents trigger runaway decoding. Our review corpus is the on-disk index of cap-hit generations, `review_corpus/00_INDEX.csv`, which holds 60 unique cap-hit document IDs. After excluding the 2 boundary cases (EOS emitted exactly at the cap), 1 false positive (legitimate long content), and 1 unclassified row, the corpus comprises 56 confirmed infinite-generation positives spanning 12 populated surface classes. We report the counts the shipped index reproduces, rather than a higher headline count that the underlying corpus index does not support.

We also disclose a device-induced shrink in the stable positive set. When we re-screened a 123-document set on a CUDA GPU rather than on an Apple-silicon (MPS) backend, the strict CUDA-positive hit rate was 6 of 123, or 4.88%. Of the MPS-positive disagreements, 8 flipped to CUDA-control, and 0 new CUDA-only positives appeared. In other words, the cross-backend non-determinism shrank the stable positive set rather than enlarging it; we treat the smaller, device-consistent CUDA set as the conservative ground truth.

2.3 A domain-conditional trigger gradient

Halt-failure is not uniformly distributed across document domains. In the review corpus, the cap-hit positives concentrate sharply in one source domain: of the indexed cap-hit documents, 38 originate from the arXiv-summarization source, against only 8 from the EDGAR (SEC filing) corpus and 1 from DocLayNet, with the remainder spread across DocVQA, FUNSD, and invoice/receipt sources. Densely-structured scientific tables and LaTeX-bearing documents trigger runaway decoding far more often than sparse filings or layout-analysis pages. From a deployment standpoint this is actionable: the trigger rate is domain-conditional, so a monitor budget can be concentrated on the high-risk document classes rather than applied uniformly. We report this as a corpus-level trigger gradient rather than a controlled rate comparison, since the source corpora differ in size and content density.

3 Method

We use a small set of instruments, all pre-registered with falsification criteria before the validating runs. The methods here are the minimum needed to support the monitor; the deeper represen-

tational probing (per-class halt-direction geometry, block contribution decomposition, crosscoder comparison) is reported in the companion representational-geometry paper.

Early-detection probing. At a fixed early generation position we fit a linear probe on the residual stream to separate cap-hit-bound generations from clean controls, sweeping the 10 cached decoder layers (L0, L4, L8, L12, L16, L20, L24, L28, L32, L35) and several generation positions, under leave-one-document-out cross-validation. This tells us *where and when* a runaway is already decidable, which is the monitor’s read site.

Logit-lens triangulation. At every cached layer we apply the final RMSNorm and the unembedding to the residual stream and read off the EOS-token logit, comparing cap-hit positives against clean controls. This localizes *where in depth* a should-halt signal becomes legible at the unembedding [6], which is the minimal depth fact behind the monitor’s read site.

Perturbation tests with pre-registered falsification. Each causal test in Section 6 carries a numeric PASS threshold fixed before the run (for example, the L0 sufficiency test pre-registers a ≥ 40 percentage-point cap-hit increase as the PASS criterion), so a failure to install or abolish halting is a recorded null against a committed bar rather than a post-hoc reading. Where a probe-significance figure is cited it uses a block-shuffle null at multiple block sizes ($B = 5, 10, 20$) with Bonferroni correction, the conservative $B = 20$ block size being the load-bearing one for autocorrelation.

4 Where to monitor: the runaway is committed early

The first deployment question is *where and when* a monitor should read. The cap-hit outcome turns out to be decidable far earlier than the visible regime change. The visible runaway transition for these documents sits around generation position ~ 600 , but a pre-registered early-detection probe shows the cap-hit decision is already *linearly detectable in the first tens of generated tokens*. Fitting the cap-hit-versus-control probe on the residual stream at a fixed early generation position and sweeping all ten cached layers, the separation is decodable as early as generation position ~ 50 , with a best leave-one-out AUC of 0.857 at layer 24 (pre-registered verdict PASS, “sequence-initiation detectable”; $N = 16$, 7 positives and 9 controls). The same probe degrades as generation position grows past the early window: at layer 24 the AUC falls from 0.857 at position 50 to 0.735 at position 600. The early positions, not the late visible transition, carry the cleanest commitment signal.

The production implication is direct. A monitor that waits for the visible ~ 600 transition reads too late, because by then the runaway is the downstream consequence of a commitment that was already made in the first ~ 50 generated tokens. A monitor reading a layer-24 readout over the first tens of tokens has a decodable early signal to act on, which is exactly the read site our runtime monitor uses (Section 8). CAVEAT: this is a single best-layer AUC at a modest sample ($N = 16$ at the early position), so we report the early-commitment timing as detectable-and-pre-registered rather

than as a precisely localized commitment point, and the layer-24 readout shares a repetition-density confound that we disclose with the readout direction.

5 The deficit is a missing halt signal

We now assemble the minimal diagnostic facts a practitioner needs to justify the monitor, all of which point to the same reading: the halt-failure state has a normal residual and a should-halt signal that exists but is mis-computed or arrives too late. The deficit is a missing halt signal, not a corrupted state and not narrowly-miscalibrated EOS.

Mechanism (companion paper). The representational geometry that explains *why* these per-class halt signals are organized the way they are is the subject of a companion representational-geometry paper (repo `halt-failure-mechanism`). That paper reports per-class halt directions that are linearly decodable and content-decoupled, a build-versus-read offset across the decoder, an opposite-sign EOS bifurcation across the two dominant surface families, a running-count residual manifold, a base-versus-fine-tune PCA/crosscoder signature showing the geometry is fine-tune-introduced, and the attractor reading those results organize. We do not reproduce those results or their effect sizes here; the present paper keeps only the behavioral facts that bear directly on the monitor.

5.1 The downstream halt signal exists but arrives too late

Logit-lens analysis shows the network *does* compute a should-halt signal, just late. The EOS logit gap between positives and controls is most negative at layer $24 = -0.419$, then flips positive to $+0.481$ at layer 32 and stays positive at $+0.336$ at layer 35 ($N = 4$ positives and 9 controls; Figure 1). The should-halt signal becomes legible at the unembedding only in the final layers, after the continuation state has been committed upstream. We carry this as the minimal depth fact behind the monitor’s read site, not as a claim about halt-direction geometry.

5.2 EOS suppression is relative, not absolute

A naive reading would be that the model *crushes* EOS to a near-zero logit. The data say otherwise. Per-step analysis of a single representative cap-hit document (`docvqa_srgb0228_p2`, traced to a 3000-token cap rather than the paper-wide 12,000-token cap) shows the chosen-token-minus-EOS raw logit margin has mean roughly 20.61 and median roughly 20.63 over the logged steps, with $P(\text{EOS})$ held to a mean of roughly 10^{-7} , ranging from about 10^{-5} at its highest down to 10^{-16} at its lowest. Strikingly, the cap-hit document’s mean EOS logit ($+10.76$) is actually *higher* than a single clean control’s ($+10.07$, `docvqa_fhxn0226_p2`, which halts on its own at 193 tokens). EOS is not absolutely suppressed; it is out-competed. As with the attention signal below, this is an $N = 1$ -vs- $N = 1$ trace and we read it as illustrative rather than statistically established. Continuation tokens simply out-score it by about 20 logits, which is enough to crush the EOS probability to a mean of roughly 10^{-7} (ranging from about 10^{-5} down to 10^{-16} over the logged steps) while the EOS logit itself stays comfortably positive (Figure 2).

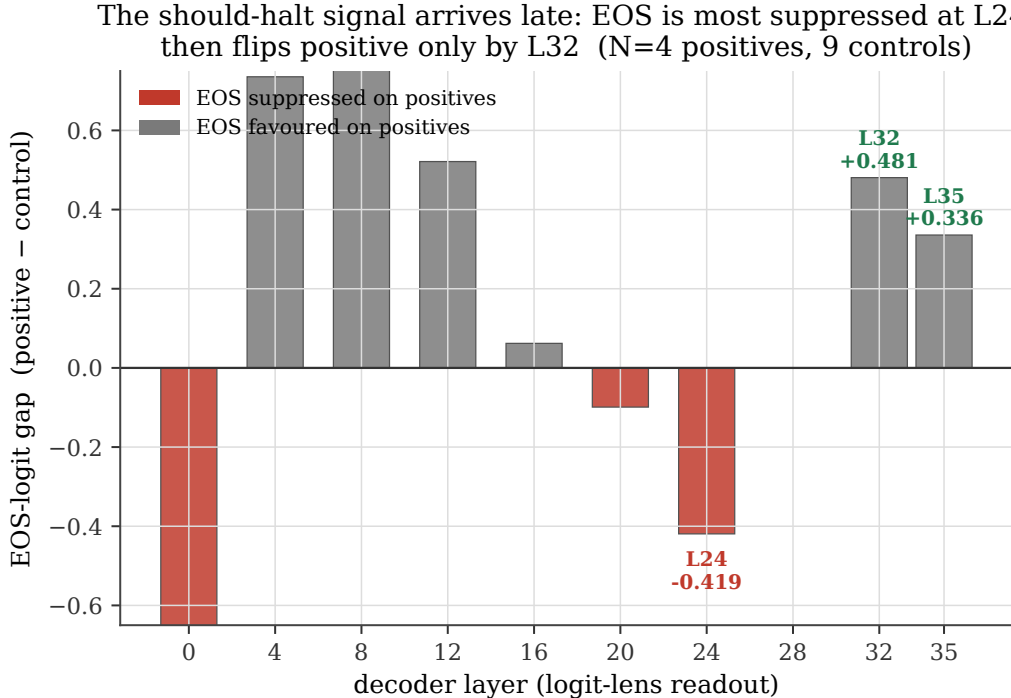


Figure 1: Logit-lens EOS-logit gap (positive minus control) by decoder layer. The gap is most negative at L24 (-0.419), meaning EOS is most suppressed there, then flips positive to $+0.481$ at L32 and $+0.336$ at L35. The should-halt signal becomes legible at the unembedding only in the final layers, after the continuation state is committed upstream ($N = 4$ positives, 9 controls).

5.3 Calibration is refuted: EOS is genuinely missing at loop onset

If the failure were a near-miss in calibration, EOS would sit just below the argmax at loop onset. It does not. The pooled record-level pool is $N = 110$ loop-onset records, but these are five contiguous records per document and are therefore pseudoreplicated; we report the statistic at the document level (effective $N = 22$ documents). Per document the EOS token sits at a median rank of roughly 12,234, no document has its per-document median rank inside the top 5, and only 1 of the 22 documents puts EOS in the top 5 at *any* logged onset position (the pooled record-level figures are a median rank of roughly 10,502 and 0.9% of records in the top 5). EOS is genuinely buried, not narrowly under-promoted. We note a framing caveat: programmatic n-gram loop-onset detection biases late, since records sit thousands of tokens deep, so “at loop onset” is a late-onset operationalization.

5.4 Coherent, not corrupted: the loop runs on a normal-norm residual

A leading competing account of degenerate generation is that the loop rides a *corrupted* residual state: an attention-sink / extreme-token regime in which a few tokens accumulate anomalously large activations and the residual norm blows up [40, 42]. That mechanism makes a sharp, falsifiable prediction about the residual L2 norm: once the loop is entered, the in-loop residual norm should

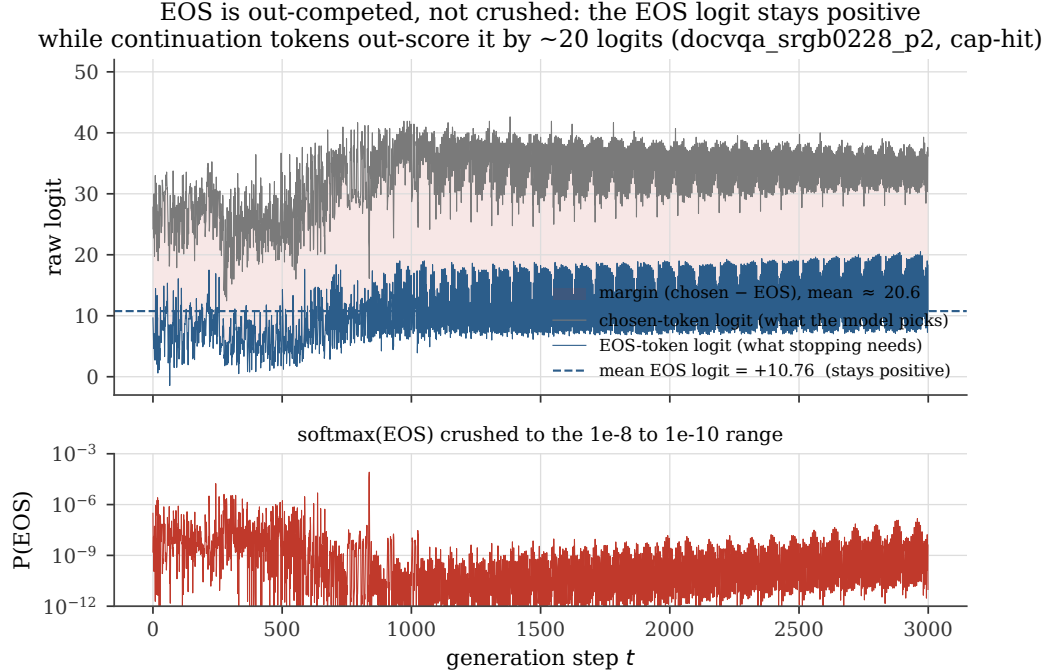


Figure 2: EOS suppression is relative, not absolute (single cap-hit document docvqa_srgb0228_p2, 3000-token cap). The EOS logit stays positive throughout (mean +10.76, which actually exceeds a single clean control’s +10.07), while continuation tokens out-score it by a mean margin of roughly 20.6 logits, crushing softmax(EOS) to a mean of roughly 10^{-7} (ranging from about 10^{-5} down to 10^{-16}). This is an $N = 1$ -vs- $N = 1$ trace, shown as an illustration of the relative-suppression mechanism rather than as a powered comparison.

be *much larger* than the pre-loop norm, with extreme-token reports placing the inflation at $\gtrsim 3\times$ the surrounding-token norm. We measure the residual L2 norm at the target layer (L24) inside versus outside the loop for the three cap-hit documents and find the opposite: the loop runs on a residual that looks *normal*, not corrupted. The in-loop/pre-loop mean residual-norm ratio is $0.96\times$ (kshm0227_p6: mean norm 86.6 in-loop vs. 90.0 pre-loop), $0.90\times$ (srgb0228_p2: 84.0 vs. 93.7), and $0.97\times$ (pgjw0227_p5: 84.3 vs. 87.3). In every case the norm *shrinks slightly* on entering the loop, and all three sit far below the $\geq 3.0\times$ inflation the attention-sink / extreme-token corruption mechanism requires. The residual norm does not merely fail to grow with the halt score; it *anti-correlates* with it: the full-trace correlation between residual norm and the halt-direction projection is negative for all three documents (-0.24 , -0.54 , -0.30). A larger norm is associated with a *more-halt* reading, not the runaway, never-halt inflation a sink-corruption account would predict.

We read this as an affirmative null that rules out the attention-sink / extreme-token corruption mechanism for our setting, rather than merely declining to find it. The loop is not running on a blown-up, sink-dominated residual that has lost the information needed to halt; it is running on a normal-looking residual whose halt-relevant projection is miscomputed. The failure is a *miscomputed decision on a coherent state*, not a *corrupted state*. This complements the relative-suppression and burial findings above (the EOS logit stays positive, just out-scored) and converts the attention-sink account from an untested competing hypothesis, mentioned only in passing in the related work,

into one we can affirmatively set aside here on the residual-norm evidence. CAVEAT: this is an $N = 3$ measurement, and the readout direction is itself repetition-density-correlated (the modal per-document verdict is “repetition-driven”: the full-trace correlation between repetition density and the halt projection is $+0.84$, $+0.90$, $+0.74$), so the anti-correlation with norm is consistent with, and does not escape, the repetition-density confound disclosed elsewhere in this paper. The norm result discriminates against *corruption*; it does not by itself identify what the halt projection *is* tracking.

6 Single-site fixes do not install halting: a converging series of nulls

If the deficit were a single localizable fault, the cheap fix would be to perturb one site: install or abolish halting by adding, removing, or zeroing a single direction, component, head, or layer. We tried this six ways and it does not work. Across six converging perturbation experiments (single-direction project-out, component-resolved patching, induction-head zeroing, prompt-engineering counterfactuals, SAE-feature ablation, and multi-layer coordinated patching), all returning the pre-registered FAIL/null verdict, no single-site intervention moves the token count. This is the systems-relevant conclusion of the paper: you cannot patch one site to make it stop. We flag at the outset what the nulls do and do not establish. They establish that single- and multi-site perturbation does not install halting at this protocol’s power. They do not establish *why*: a converging set of nulls is equally consistent with a genuinely distributed deficit and with a protocol under-powered at the halt-relevant locus, especially since three of the six rest on only $N = 2$ documents. We do not adjudicate that mechanistic question here (Section 10); the present section reports the nulls and the one positive control that bounds, but does not eliminate, the under-powered alternative.

6.1 Sufficiency null at L0

Injecting the class-matched bare-word-repeat halt direction at layer 0 does not induce halt-failure. At the pre-registered target scale ($\alpha = 10$), the change in cap-hit rate versus the vanilla run is 0.0 percentage points, against a pre-registered PASS threshold of ≥ 40 pp, a clear failure of the sufficiency hypothesis. Layer 0 is a readout, not a generator.

6.2 Norm-scaled positive control bounds, but does not eliminate, “protocol under-powered”

A reviewer might object that the perturbation protocol is simply too weak to move anything. The norm-scaled positive control speaks to this, but only partially, and we state its composition precisely because it is load-bearing. The run scaled a single-direction layer-24 patch to $0.1\times$, $10\times$, and $100\times$ the residual norm over four documents, but only *three* of those were cap-hit positives (manifest label **positive**: vanilla runs of 12,000 tokens). On those three positives, none escapes the cap at any scale; all stay pinned at 12,000. The perturbations nonetheless *do* land visibly in the output (one document gets a foreign-language token prepended, another gets stray Markdown emphasis inserted), so the protocol is demonstrably not inert: it delivers content-altering perturbations; it

just cannot dislodge the halt-failure state. Content moves while token count does not. The fourth document (`table_N005_C04_s05000`, manifest label `control`) was *not* a cap-hit positive (it halts cleanly on its own at 477 tokens with an EOS stop), and so it cannot “escape” a cap it never hit. We flag it explicitly rather than fold it into the positive tally, because under the same perturbation its output *collapsed*: to 3 tokens at $10\times$, to 1 token at $0.1\times$, and to 12,000 tokens of degenerate single-character repetition at $100\times$. That collapse is a norm-shock signature; if anything it *weakens* the protocol-sensitivity rebuttal, by showing that this matched-norm patch can shatter a clean halt rather than cleanly perturb it, so we do not count it as a positive. We are careful about what the three positives rule out. They rebut the *strongest* form of the under-powered objection (that the protocol delivers no perturbation at all) but they do not rebut the weaker, more relevant form: that the protocol cannot perturb the *specific* halt-relevant direction at the *specific* halt-relevant locus. A random-direction patch that visibly reorders surface tokens (or shatters an unrelated clean halt) is not evidence that a targeted patch of the halt mechanism would have been detectable. We therefore treat the positive control as a sensitivity floor, not as a clean refutation of under-power.

6.3 The on-record null count, stated honestly

Our strongest necessity test is a reverse-direction project-out experiment: it asks whether *removing* the halt direction, by projecting it out of the residual stream at every position, lengthens generation. It does not. Across three clean-halting controls the projected-out run is unchanged (within 10%: +9.1%, 0%, 0%), and the positive document stays pinned at the cap, yielding a readout verdict. Importantly, the project-out hook is confirmed to fire: two of the four documents produce materially different output under it (one positive flips its table from Markdown to HTML), so this is a genuine null and not a silently inert intervention (Figure 3).

A component-resolved patch grid points the same way. It decomposes the patch into MLP, attention, and block contributions at layers 16, 20, and 24, across 5 documents and 4 intervention types: 0 of 45 real-patch cells produce any length reduction (a clean real-null), while 9 of 134 control cells do (Figure 4). The honest, pre-registered caveat is sharper than off-layer norm-shock alone: only 4 of the 9 nonzero controls are off-layer perturbations, while the remaining 5 fire *at the same layer and component as the real patch* (2 random-matched-norm, 3 shuffled-values). The clearest warning sign is at L16/block_output, where a random vector matched to the real patch’s norm (≈ 47.4) reduced one document’s length by 99.95% while the real readout-direction patch at exactly that site reduced it by 0%. This makes “the controls themselves are suspect” (the protocol’s own perturbations can collapse generation in ways unrelated to any halt direction) a live alternative reading, not merely off-layer norm-shock, so the grid bounds protocol sensitivity rather than localizing a mechanism. The verified positive control above is our best evidence that the protocol has real sensitivity.

Accordingly, our load-bearing causal evidence is *six converging perturbation nulls*, all returning the pre-registered FAIL/null verdict, with N ranging from 2 to 22 documents. We are precise about provenance: the two anchor nulls (B3 and P16) are verified on disk in this repository; the other four (P10b, P19, P12, and P22) are recorded with their pre-registered FAIL verdicts and their primary summary files are pending re-sync for external release. The six are:

1. **B3, reverse-direction project-out null** (necessity; READOUT verdict; 3/3 controls unchanged, positive pinned at the cap, project-out hook confirmed to fire;

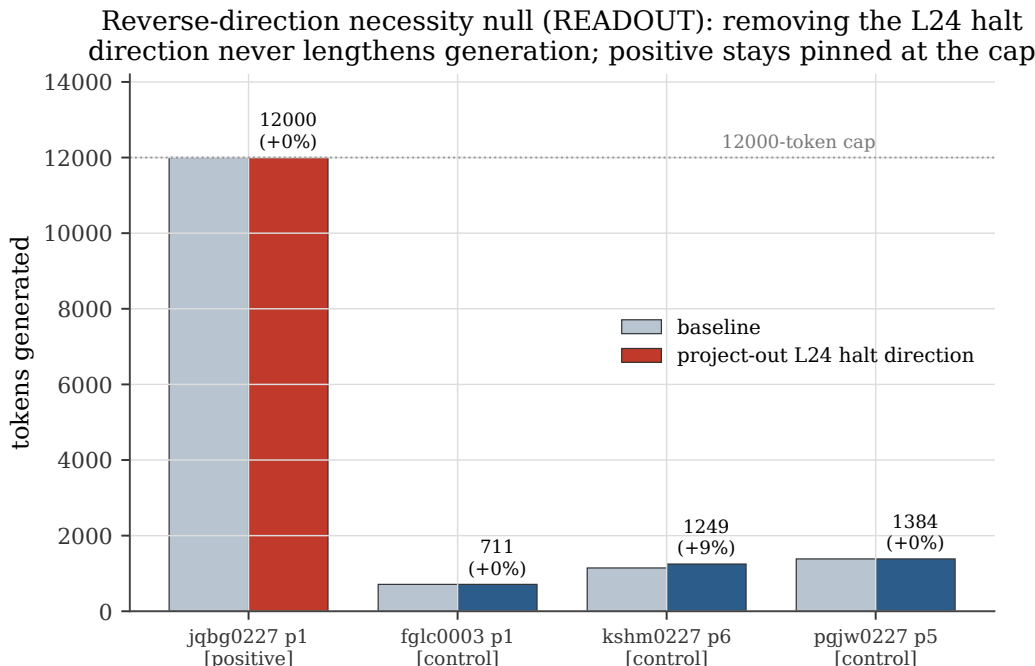


Figure 3: Reverse-direction necessity null (READOUT verdict). Projecting out the L24 halt direction at every position does not lengthen generation: the positive document stays pinned at the 12000-token cap (12000 to 12000, +0%) and the three clean-halting controls are unchanged (+9.1%, 0%, 0%). The project-out hook is confirmed to fire (output content changes without length change), so this is a genuine null.

p2_pilot/b3_reverse_direction_summary.json);

2. **P16, component-resolved patch grid** (necessity; FAIL verdict; 0/45 real-patch cells reduce length, 9/134 control cells do, $N = 5$ documents; p16_component_resolved_short/_summary.json). P16 subsumes the earlier component claims CL-19 and CL-20 and is counted once here, not as two separate nulls;
3. **P10b, induction-head zeroing** (necessity; FAIL verdict; 0/4 induction heads pass the pre-registered 30% threshold under zeroing, $N = 2$ documents; p10b_head_zero_patching/_summary.json);
4. **P19, system-prompt counterfactual** (FAIL verdict; 0/6 stop-instruction prompts pass, median reduction 0%, $N = 5$ documents; p19_system_prompt_counterfactual/_summary.json);
5. **P12, SAE-feature ablation** (FAIL verdict; 0/5 cross-validation folds pass all pre-registered criteria, top-feature AUC 0.54 to 0.63 against a 0.85 bar, $N = 22$ positives versus 56 controls; p12_sae_ablation/_summary.json);
6. **P22, multi-layer coordinated patch** (FAIL verdict; the real coordinated zero-patch of the L16/L20/L24 window reduces length 0%, $N = 2$ documents; p22_multilayer_coordinated/_summary.json).

Component-resolved patch grid: the real patch never moves length.
0/45 real cells vs 9/134 control cells (verdict FAIL)

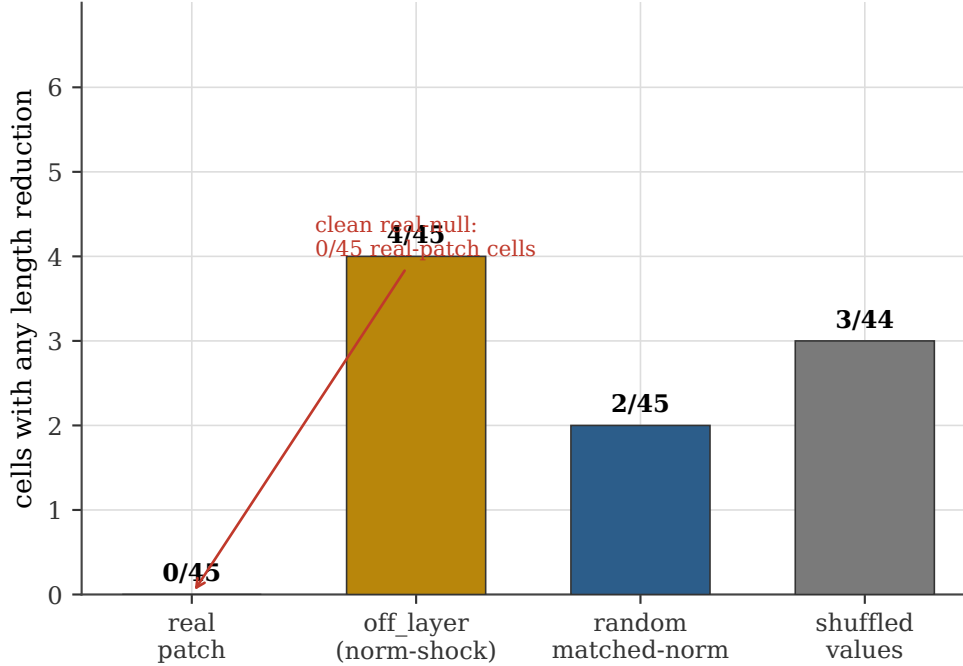


Figure 4: Component-resolved patch grid by intervention type. The real patch never moves length (0/45 cells, a clean real-null), while 9/134 control cells show a nonzero reduction: 4 off-layer, 2 random-matched-norm, and 3 shuffled-values. Only 4 are off-layer; the other 5 fire at the same layer and component as the real patch. Most starkly, at L16/block_output a random vector matched to the real patch’s norm (≈ 47.4) cuts one document’s length by 99.95% while the real patch there cuts 0%. Verdict FAIL. These nonzero controls make “the controls are suspect” a live reading and bound protocol sensitivity rather than localizing a mechanism.

These six are independent perturbation classes (project-out of a single direction, forward-patching of resolved components, head-level zeroing, prompt-level counterfactuals, SAE-feature ablation, and a multi-layer coordinated patch), and each carries its own positive or control evidence, so they are not a single test reported six times. We keep the count honest in both directions. On provenance: B3 and P16 are verified on disk, while P10b, P19, P12, and P22 are recorded with their FAIL verdicts and primary files pending re-sync, so the load-bearing local anchors are the first two. On power: three of the six rest on small samples ($N = 2$ for B3’s positive, P10b, and P22), and a uniform null at small N does not by itself separate a genuinely distributed deficit from an under-powered protocol (Section 10). We also carry the controls-are-suspect caveat openly: in the component grid, 5 of the 9 nonzero controls fire at the same layer and component as the real patch (not merely off-layer), and a norm-matched random vector (≈ 47.4 , the real patch’s norm) reduced one document’s length by 99.95% at L16/block_output where the real readout-direction patch reduced 0%. That a matched-norm random perturbation collapses generation where the real patch does nothing makes “the controls themselves are suspect” a live alternative to “the protocol is under-powered,” rather than a simple off-layer norm-shock artifact.

We state the bottom line for this section plainly, because it is the load-bearing systems conclusion of the paper. All single- and multi-site perturbations we ran returned null at this protocol’s power: you cannot patch one site to make the model stop, so the deficit is not a single steerable fault. Whether the underlying deficit is genuinely distributed or the protocol is simply under-powered at the halt-relevant locus is *not resolved* by these data, and we do not need to resolve it for the systems claim. The discriminating experiment (a positive-control patch of the *same* readout direction at a *non-loop* position, which would separate “distributed” from “under-powered”) is left to the companion representational-geometry paper and to future work.

7 Cross-family: motivation only

To motivate that this is not one fine-tune’s idiosyncratic bug, we ran a small 4×4 document-by-model grid spanning four model families (SmolDocling-256M-preview, InternVL3-8B, Qwen3-VL-8B-Instruct, and Qwen3-VL-30B-A3B-Instruct) crossed with four documents. Under a strict criterion (`stop_reason == max_new_tokens`), halt-failure reproduces on 10 of the 16 cells, with at least one cap-hit cell in every family (per-family 3/4, 1/4, 3/4, 3/4); a more permissive near-cap relabel raises the tally to 11/16, so the count is criterion-dependent. Sixteen cells over four families is far too thin to support a generality claim, and the directory-level aggregate verdict over this run abstains (`INSUFFICIENT_DATA`). We therefore use this only as a motivation that the phenomenon is not unique to the studied fine-tune, and defer any actual generality claim to a dedicated companion note that owns the cross-family matrix.

8 The deployable artifact: a runtime monitor and its honest trade-off

The deliverable is a runtime monitor that reads the layer-24 readout direction at the early site identified in Section 4 and boosts the EOS logit when the readout fires. It suppresses runaway generation, but it sits on a genuine precision-recall trade-off rather than a free lunch, and there is no free sweet spot.

At a +6.91 EOS boost, the reactive monitor achieves a 37.68% mean length reduction with 0 of 10 control false-positives and 5 of 13 positive cap-hits remaining (Figure 5). This is the deployable clean-specificity operating point. At a +30 EOS boost, the same monitor achieves a 99.58% mean length reduction with 0 remaining cap-hits, but at a catastrophic 10 of 10 control false-positives. The two operating points bracket the trade-off: you can buy near-total loop suppression only by destroying clean-document specificity, so the deployable point is the conservative +6.91 boost. We are explicit about the scope of the false-positive bound: the 0-of-10 figure is measured on $N = 10$ clean controls, a workshop-scale specificity check, not a large-corpus false-positive rate. Characterizing the false-positive rate at the $N \geq 600$ long-form scale a production deployment would require is left to future work; the present claim is a small- N specificity bound, not a deployment-grade FP guarantee.

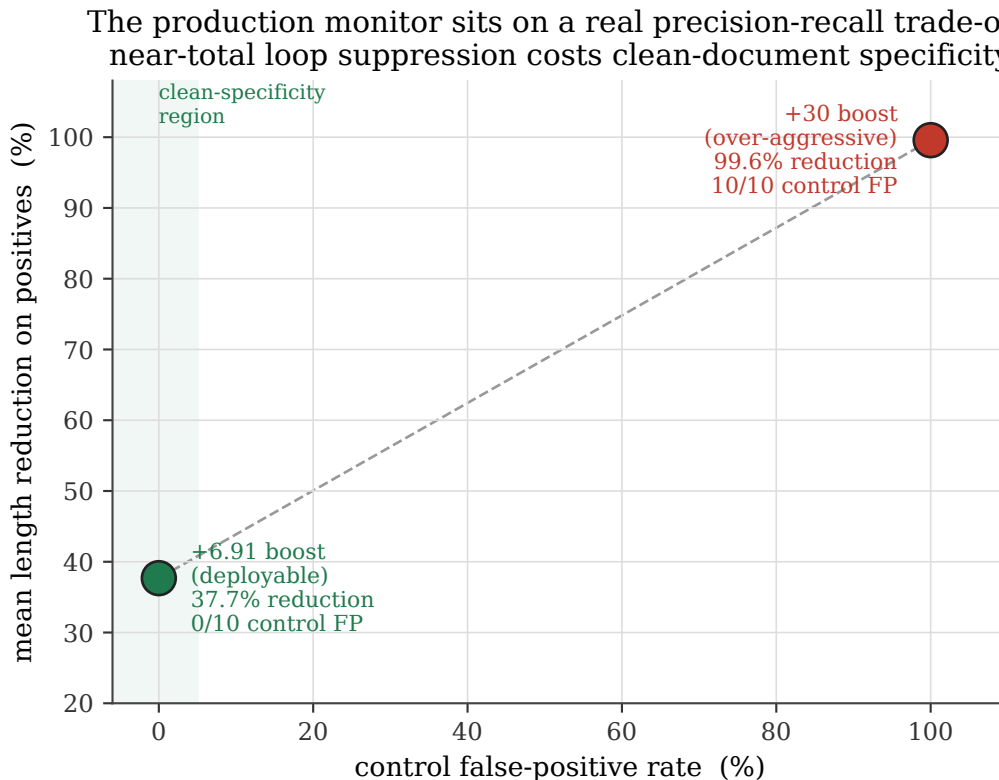


Figure 5: The production monitor sits on a real precision-recall trade-off. At a +6.91 EOS boost it gives 37.7% mean length reduction with 0/10 control false-positives (deployable clean-specificity point); at a +30 boost it gives 99.6% reduction with 0 remaining cap-hits but 10/10 control false-positives. Near-total loop suppression costs clean-document specificity. $N = 13$ positives, 10 controls.

Image content is sufficient to sustain the loop. Replacing the input image with grey noise collapses every tested cap-hit positive from a full-cap runaway to a short clean-EOS halt: across the tested positives, baseline cap-length stops become 13- to 18-token, EOS-terminated halts under grey-noise inpaint. All 5 of 5 explicitly named positives collapse this way (`arxiv_table_000266`, `docvqa_jqbg0227_p1`, `docvqa_kshm0227_p7`, `docvqa_srgb0228_p2`, `funsd_105_105`); with $N = 5$ we report the count qualitatively rather than attaching a Wilson interval. We state the claim as image *content* being sufficient to sustain the loop rather than as a clean vision-causality demonstration, because grey-noise replacement confounds removing visual evidence with destroying readable content; the discriminating experiment (swapping in a different readable document) is left to future work. This still motivates a vision-aware guardrail rather than a purely text-side penalty: the halt-failure state is conditioned on the visual input.

A correlational vision signal points the same way. In a single-positive-versus-single-control comparison, the positive document’s text-to-image attention mass collapses in the L20 to L24 halt band: the positive-to-control ratio is 0.070 at layer 20 ($14.2\times$ lower), 0.216 at layer 24 ($4.6\times$ lower), and only a modest 0.695 at layer 16. The collapse is band-localized rather than monotone: the *same* document pair shows the positive with *higher* image attention than the control at layer 8

(1.10 $\times$) and again at layer 35 (1.37 $\times$), so the L20-to-L24 dip is suggestive only. We flag this as $N = 1$ -vs- $N = 1$ and reserve it for future larger-sample work.

9 Related work

Our work sits at the intersection of seven threads: language-model degeneration, hallucination in vision-language models (VLMs), the mechanistic-interpretability toolkit, single-direction steering, the geometry of count-and-halt decisions, attention-sink accounts of termination, and cross-modal grounding collapse. We review each and locate our contribution against it.

Degeneration, repetition, and non-termination. Failure to terminate is a long-standing pathology of likelihood-trained autoregressive decoders: likelihood decoding produces bland, repetitive, degenerate text [14], and greedy, beam, top- k , and nucleus decoding are all provably inconsistent, able to emit infinite-length, zero-probability sequences unless the model is made self-terminating [38]. The training objective has itself been implicated, motivating unlikelihood training to suppress repetition at the source [37]. In production OCR pipelines the symptom is acute enough to be engineered around directly: retry mechanisms and post-hoc filters are now standard [27, 28], and repetition is documented as a first-class production concern in both general LLM serving [35] and recent VLM releases [36]. These works treat non-termination as a symptom to be suppressed at the decoding or serving layer rather than as a representational phenomenon to be localized. A complementary, mechanistically-flavored line asks *why* the decoder loops. The repeated-token phenomenon has been interpreted at the level of early-layer attention and sink structure [42], and single-neuron and single-head accounts of repetitive generation exist for text-only models: “repetition neurons” identify neuron-level circuits that drive repeats [13], while induction-head accounts trace pattern completion to match-and-copy heads [25] and quantify when those heads over-fire on repeated context [34] (note: this preprint was withdrawn in December 2025). These are the closest causal hypotheses to ours, and we situate against them deliberately. They are perpendicular on three axes: they study text-only LLMs rather than fine-tuned OCR-VLMs, they target repetition as such rather than the broader *halt-failure* phenomenon (of which token loops are one surface class among several, including phantom table rows and structural-marker hallucination), and they localize repetition to single neurons or heads (and, where intervened, report that ablating or regularizing them changes behavior). Our central systems finding is the opposite: a converging series of single-direction and single-layer perturbations *fails* to move the token count, which is why we conclude that single-site fixes do not install halting (whether the underlying deficit is genuinely distributed is taken up in the companion paper).

Hallucination in (V)LMs, including structural and OCR hallucination. Most VLM hallucination research concerns object hallucination in captioning and VQA, as captured by benchmarks such as POPE [20] and surveyed broadly in Bai et al. [5]. Document OCR introduces a distinct, *structural* failure mode: phantom blank table rows, fabricated page numbers and watermarks, and runaway markup loops; the closest mitigation-side work studies OCR hallucination under degraded input on the same model family with a dedicated benchmark and an uncertainty-aware fix [12]. We study this on a fine-tune of the Qwen2.5-VL family [4]; the same surface symptoms recur

across OCR-VLM releases [15, 27, 28, 36]. The mechanistic substrate for OCR-specific behavior has begun to be mapped: text-inpaint causal protocols localize where OCR information enters the residual stream [30], and path-patching studies isolate vision-to-text “OCR heads” that route visual evidence into the language stream [3]. We borrow the text-inpaint logic for our image-content test, where replacing the input image with grey noise collapses tested cap-hit positives (all 5 of 5 named documents) from 12,000-token runaways into short `stop_reason=eos` halts of 13 to 18 tokens, establishing that image content is sufficient to sustain the halt-failure state (we say “sufficient” rather than “causally load-bearing” because grey noise confounds removing visual evidence with destroying readable content). Where these works localize *where OCR content arrives*, we ask *where and when the runaway becomes detectable* and show that the decision to terminate is missing rather than mislocated at a single routing site.

Mechanistic-interpretability methods. Our diagnostics are a small subset of the standard mechinterp toolkit. We fit linear probes [1] for early detection of the cap-hit outcome, and we triangulate the depth at which a “should-halt” signal becomes legible at the unembedding using the logit lens [24] and its better-calibrated tuned-lens variant [6]; this is what surfaces our minimal depth fact that the EOS logit gap is most negative at L24 and only flips positive by L32, i.e., the should-halt signal arrives too late to override an already-committed continuation state. Our perturbation tests use activation and path patching in the lineage of causal-mediation and circuit-level analysis [22, 33], including component-resolved causal tracing [19] and SAE-feature ablation [7, 8, 26], all implemented as direct PyTorch forward hooks following the intervention-API conventions of [10, 39]. The deeper representational diagnostics that these methods also enable (per-class halt-direction geometry, build-versus-read decomposition, and a base-versus-fine-tune crosscoder comparison [23] showing the geometry is fine-tune-introduced) are reported in the companion representational-geometry paper rather than here. The “readout-versus-locus” distinction we draw at L24 shares its mechanism vocabulary with hallucination-detection work that reads model internals [31].

Single-direction and steering methods. The hypothesis that a behavior lives along a single residual-stream direction, testable by adding or projecting out that direction, is most cleanly stated for refusal, which is mediated by a single direction recoverable and steerable across many chat models [2], and has a direct method-class ancestor in inference-time intervention, which steers along an attention-head truthfulness direction [18]; the underlying add-a-difference-vector mechanism is the activation-engineering recipe our injections instantiate [32]. That single-direction picture is itself contested: refusal has been shown to factor into geometrically distinct, class-specific directions rather than one universal steerable axis [16], a class-specific picture the companion paper develops for halt directions. Our production monitor is the OCR-halt analogue of these methods: it reads an L24 readout direction and boosts the EOS logit rather than shifting an activation. The closest VLM-side comparable performs decoder-time steering pre-decoding rather than via a mid-decoder forward hook [29], and class-conditional steering offers a finer-grained alternative when one global direction is too coarse [41]. Crucially, our single-direction tests come back null. Injecting a class-matched halt direction at L0 changes the cap-hit rate by 0.0 percentage points against a pre-registered 40-point PASS threshold, and a norm-scaled positive control fails to dislodge the halt state at 0.1 $\times$, 10 $\times$, or 100 $\times$ the residual norm (0 of 3 cap-hit positives escape the cap at any

scale; a fourth, clean-halt control in the same run instead collapsed under the patch) even though the perturbations visibly alter output content. This is the inverse of the steering-success narrative in the single-direction literature: the readout is *decodable* without being *steerable*. We read this decodable-but-not-steerable gap as the systems statement that single-site fixes do not install halting, while noting it is equally consistent with a protocol under-powered at the halt-relevant locus; we do not adjudicate between the two (Section 10).

The counting-manifold account of halt decisions. The decision to stop is, in part, a decision about *how much* has been emitted, which connects halt-failure to the geometry of counting, where a running count is tracked along a low-dimensional curved manifold and used to make a stop-or-continue decision [11]; the companion representational-geometry paper takes up whether a count-manifold and build-versus-read offset organize the halt signals, which is out of scope for this systems paper.

Attention sinks, EOS-halting, and SinkTrack. A prominent family of accounts ties termination and long-context stability to attention sinks. The attention-sink phenomenon was first identified on the initial tokens of a sequence and exploited for stable infinite-length streaming generation [40]. Building on it, SinkTrack treats an anchor token as an information sink and injects context into it to counteract attention drift, reporting multimodal gains [21], and the repeated-token failure has itself been traced to attention-sink structure [42]. This is the most actionable competing mechanism for our setting and a natural baseline. Our evidence is only partly aligned with it: we find that EOS suppression is *relative, not absolute*, with continuation tokens out-scoring EOS by roughly 20 logits while the EOS logit itself stays positive (in a single $N = 1$ -vs- $N = 1$ trace a cap-hit document’s mean EOS logit of +10.76 exceeds a clean control’s +10.07), and that at loop onset EOS is genuinely buried (per-document median rank near 12,000 over 22 documents, only 1 of which ever puts EOS in the top 5), rather than narrowly under-promoted by a single mis-set sink. More directly, we test the residual-norm prediction this account makes, a $\gtrsim 3\times$ in-loop norm inflation, and find the opposite (Section 5): the in-loop/pre-loop residual norm is $0.96\times/0.90\times/0.97\times$ across the three cap-hit documents, far below the $\geq 3.0\times$ corruption regime, and the norm anti-correlates with the halt score. The loop runs on a normal-norm, coherent residual, so we read the failure as a miscomputed decision rather than a sink-corrupted state, and we affirmatively set the attention-sink / extreme-token corruption mechanism aside for our setting (at $N = 3$, with the repetition-density confound noted). A sink-anchoring intervention is therefore a complement to, not a substitute for, the missing-halt-signal reading.

Cross-modal grounding collapse. Finally, several works frame VLM hallucination as a drift of attention away from visual evidence toward linguistic priors. Visual contrastive decoding contrasts outputs from original versus distorted images to suppress over-reliance on language priors [17], and a complementary line reorients attention toward image tokens at the moment grounding is needed: confidence-aware attention calibration rescales attention to counter late-generation modality drift [9], and attention-lens analyses diagnose text-vs-vision routing as the locus of modality bias and re-balance the two streams [43]. This “detect the decision moment, then intervene” structure matches our production setting, and the modality-bias account is corroborated on our side by a (single-

positive-versus-single-control, hence preliminary) collapse of text-to-image attention mass in the L20 to L24 band.

Positioning. Against all of these, our systems contribution is to read OCR-VLM halt-failure not as a single broken layer, sink, head, or steerable direction, but as a *missing halt signal*: a should-halt signal that exists yet arrives too late and is out-competed, running on a coherent (not corrupted) residual, robust to single-site perturbation in both the necessity and sufficiency directions, and conditioned on the visual input. On that reading we build the deliverable, a runtime readout monitor with an honest precision-recall trade-off rather than a clean single-direction fix, and we treat the converging nulls as first-class evidence that single-site fixes do not install halting. The representational geometry that organizes the per-class halt signals (and the attractor hypothesis it supports) is the subject of the companion representational-geometry paper, not this one.

10 Limitations and walk-back

We state the walk-backs plainly because they are part of the contribution.

1. **“Distributed” is not resolved against “under-powered,” and we do not need to resolve it.** All single- and multi-site perturbations we ran returned null at this protocol’s power, and whether the underlying deficit is genuinely distributed across many directions or the protocol is simply under-powered at the halt-relevant locus is not resolved by these data. The two readings are observationally equivalent at this protocol power. This is acceptable for the present paper because our claim is the systems one, that single-site fixes do not install halting, not the mechanistic one that the deficit is distributed. The experiment that would settle it (a positive-control patch of the same readout direction at a non-loop position) is left to the companion representational paper and to future work.
2. **Necessity evidence is six converging nulls, but provenance and N are uneven.** Six perturbation nulls all return the pre-registered FAIL/null verdict: the reverse-direction project-out null (B3, readout verdict, controls unchanged, positive pinned at the cap, project-out hook confirmed to fire), the component-resolved patch grid (P16, 0 of 45 real-patch reductions, $N = 5$), induction-head zeroing (P10b, 0 of 4 induction heads pass under zeroing, $N = 2$), the system-prompt counterfactual (P19, 0 of 6 stop-instruction prompts pass, $N = 5$), SAE-feature ablation (P12, 0 of 5 folds pass all criteria, top-feature AUC 0.54 to 0.63 versus a 0.85 bar, $N = 22$ positives), and a multi-layer coordinated zero-patch (P22, real coordinated patch reduces length 0%, $N = 2$). On provenance: B3 and P16 are verified on disk in this repository; P10b, P19, P12, and P22 are recorded with their FAIL verdicts but their primary summary files are pending re-sync before external release, so the two anchor nulls carry the local load. On power: three of the six rest on small samples (B3’s positive document, P10b, and P22 each at $N = 2$), and a uniform null at $N = 2$ does not on its own discriminate a distributed deficit from an under-powered protocol; that disambiguation (item 1 above) remains open. The residual caveat is sensitivity, not verification: in the P16 grid 5 of the 9 nonzero control patches fire at the same layer and component as the real patch (only 4

are off-layer), so the six nulls establish “single- and multi-site perturbation does not install halting at this protocol’s power,” not a fully exhaustive causal search.

3. **The controls are suspect, and the positive control only bounds under-power.** In the component grid, a norm-matched random vector (≈ 47.4) cut one document’s length by 99.95% at L16/block_output where the real readout-direction patch cut 0%, and 5 of the 9 nonzero controls are not off-layer. That matched-norm random perturbations can collapse generation where the real patch does nothing means “the controls are suspect” is a real alternative to “the protocol is under-powered.” Our verified positive control (Section 6) is the load-bearing sensitivity evidence, but it rebuts only the strongest form of the under-powered objection (the protocol is not inert; it visibly alters content). It does not rebut the weaker, more relevant form, that the protocol cannot perturb the specific direction at the specific halt-relevant locus, which is why item 1 above remains open.
4. **Cross-family reproduction is small-matrix and the aggregate verdict abstains.** Under the strict `stop_reason == max_new_tokens` criterion, halt-failure reproduces on 10 of 16 doc-model cells across four families (per-family 3/4, 1/4, 3/4, 3/4), but the matrix is only sixteen cells and the directory-level aggregate verdict over this run is `INSUFFICIENT_DATA`. We make no strong generality claim from it. A permissive near-cap relabel would shift the tally to 11/16, so the count is criterion-dependent. We use it as motivation only and defer a properly powered cross-family study to a companion note.
5. **The monitor and image-inpaint results are workshop- N .** The monitor trade-off rests on $N = 13$ positives and $N = 10$ controls, and the image-inpaint result on $N = 5$ named positives. The 0-of-10 false-positive figure is therefore a small- N specificity bound, not a deployment-grade false-positive rate; characterizing the FP rate at the $N \geq 600$ long-form scale a production deployment would require is future work.
6. **The vision-attention-collapse signal is $N = 1$ -vs- $N = 1$ and non-monotone.** The attention-collapse comparison is a single-positive-versus-single-control measurement, not yet statistically meaningful, and the band-localized L20-to-L24 dip is suggestive only: the same document pair shows the positive with *higher* image attention than the control at L8 (1.10 $\times$) and L35 (1.37 $\times$). It is reserved for future larger-sample work.
7. **Loop-onset framing biases late, and the calibration pool is clustered.** The calibration result is computed at programmatically detected onset positions that sit thousands of tokens deep; this overstates “onset” and is a framing concern, though it does not change the numeric finding that EOS is genuinely missing. The 110 onset records are five contiguous records per document and so are pseudoreplicated; we therefore report the statistic at the document level (effective $N = 22$), where no document has a median EOS rank inside the top 5 and only 1 of 22 puts EOS in the top 5 at any logged position.

11 Conclusion

Halt-failure in a fine-tuned OCR-VLM is best read, from a systems standpoint, as a *missing halt signal* rather than a single broken layer, a corrupted state, or mere repetition. The supporting

facts are pre-registered and converging: the network computes a downstream should-halt signal that arrives too late to override an upstream continuation state (logit-lens EOS gap -0.419 at layer 24 flipping to $+0.481$ by layer 32); that suppression is relative not absolute (chosen-vs-EOS margin about 20 logits; an $N = 1$ illustration); EOS is genuinely missing at loop onset (per-document median rank roughly 12,234 over 22 documents, only 1 of which ever puts EOS in the top 5); the loop runs on a coherent, not corrupted residual (in-loop/pre-loop norm 0.90 to $0.97\times$, ruling out the $\geq 3\times$ inflation a sink-corruption account requires; $N = 3$); and six converging single-direction, single-layer, and multi-layer perturbations fail to move the token count. On that last point we are deliberately careful: whether the underlying deficit is genuinely distributed or the protocol is under-powered at the halt-relevant locus is *not resolved* by these data, and we do not need to resolve it for the systems claim that single-site fixes do not install halting. Our load-bearing causal evidence is six converging perturbation nulls, all returning the pre-registered FAIL/null verdict (N ranging 2 to 22 documents): B3 and P16 are verified on disk, while P10b, P19, P12, and P22 are recorded with their FAIL verdicts and their primary files pending re-sync. Three of the six rest on $N = 2$, and a uniform null at small N does not by itself separate “distributed” from “under-powered.” The surface symptom is not unique to the studied fine-tune: under a strict stop-reason criterion halt-failure reproduces on 10 of 16 doc-model cells across four model families (11 of 16 under a permissive relabel), but the matrix is small and the aggregate verdict abstains, so we use it as motivation only. The deliverable is a deployable runtime readout monitor with an explicit precision-recall trade-off (37.68% mean reduction with 0 of 10 control false-positives but still leaving 5 of 13 positives at the cap, or 99.58% reduction at 10 of 10 control false-positives), reading the layer-24 readout at the early-commitment site, plus an image-inpaint result showing that image content is sufficient to sustain the loop (all 5 of 5 named positives collapse to short EOS halts under grey noise), which motivates a vision-aware guardrail. The walk-backs are part of the result: showing where the single-cause story fails is what justifies treating the deficit as a missing halt signal and shipping an honest trade-off rather than a clean single-direction fix. The representational geometry that explains *why* the per-class halt signals are organized as they are, and the attractor hypothesis it supports, is the subject of a companion representational-geometry paper.

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